

Advanced Statistical Methods II

Lecture II: Types of Missingness and Their Implications

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Outline of Today's Lecture

Recap and today's central question

MCAR

MAR

MNAR

Comparison of these three

Ignorability and observed-data likelihood

Diagnostics, tests, and limitations

Implications for estimation

Key Takeaway Messages

Incomplete data can affect the statistical methodologies.

In particular, we saw that:

- ▶ a complete-data likelihood can be easy to write;
- ▶ an observed-data likelihood may require summing or integrating over what is unobserved;
- ▶ naive deletion or naive filling-in can change both bias and uncertainty;
- ▶ the **reason for missingness** matters as much as the amount missing.

Today's guiding question



When can we safely ignore the missingness process?

It asks whether the observed part of the data is statistically representative enough for the inferential task at hand.

If two datasets have the same percentage of missing values, can one be harmless and the other seriously biased?

Running example: exam performance data



Suppose we are studying the relationship between study hours and exam score.

Student	Study hours	Attendance	Stress score	Exam score
1	4.5	82%	3	71
2	7.0	93%	2	88
3	1.5	45%	8	?
4	5.5	76%	5	79
5	2.0	50%	7	?

Possible stories:

- ▶ score missing due to a random data-entry failure;
- ▶ score missing because low-attendance students missed the exam;
- ▶ score missing because students who performed poorly refused to report it.

Same table. Different statistical implications.

Full data, observed data, missing data



Let the full data be denoted by $Y = (Y_{\text{obs}}, Y_{\text{mis}})$ where

- ▶ Y_{obs} is the part we actually observe;
- ▶ Y_{mis} is the part that exists conceptually but is not observed;
- ▶ inference is about a model parameter θ in the distribution of Y .

Observed-data likelihood

Following the incomplete-data likelihood notation of [Rubin \(1976\)](#) and [van Buuren \(2018, Chapter 2, Sections 2.2.3–2.2.4\)](#), for continuous missing values,

$$L_{\text{obs}}(\theta) = f(Y_{\text{obs}} \mid \theta) = \int f(Y_{\text{obs}}, Y_{\text{mis}} \mid \theta) dY_{\text{mis}}.$$

For discrete missing values, the integral is replaced by a sum.

The missingness indicator



For each unit and variable, define an indicator

$$R = \begin{cases} 1, & \text{if the value is observed,} \\ 0, & \text{if the value is missing.} \end{cases}$$

The missingness mechanism



In Rubin's theory every data point has some likelihood of being missing.

The process that governs these probabilities is called the *missing data mechanism* or *response mechanism*.

A missingness mechanism is a probabilistic model for the response pattern (Rubin, 1976; Little and Rubin, 2019):

$$\mathbb{P}(R \mid Y_{\text{obs}}, Y_{\text{mis}}, \psi),$$

where ψ denotes parameters governing the missingness process.

- ▶ The data model is about Y and parameter θ .
- ▶ The missingness model is about R and parameter ψ .
- ▶ The central question is how R depends on Y_{obs} and Y_{mis} .

A warning about notation



Different books use slightly different notation. In these slides, we use Rubin's response-indicator convention with $R = 1$ for observed and $R = 0$ for missing, as in the standard likelihood formulation of [Rubin \(1976\)](#); see also [van Buuren \(2018, Section 2.2.3\)](#).

Sometimes we also have fully observed covariates X , such as age, sex, treatment arm, school, or sampling stratum.

Then one may write

$$\mathbb{P}(R \mid Y_{\text{obs}}, Y_{\text{mis}}, X, \psi).$$

Whether X is considered part of Y_{obs} or written separately is a matter of notation. The key issue is whether missingness depends on information we have observed or on information we have not observed.

Three Types of Missingness



Rubin (1976) classified missing data problems into three categories:

1. MCAR
2. MAR
3. MNAR

The data are **missing completely at random** if the probability of missingness does not depend on either observed or missing values:

Intuition:

- ▶ the missing cases are like a random subsample of the full dataset;
- ▶ the missingness process is unrelated to the variables. The causes of the missing data are **unrelated** to the data.
- ▶ the observed sample remains representative, apart from random loss of information.

MCAR is a strong condition. The definition follows the response-mechanism formulation of [Rubin \(1976\)](#); see also [van Buuren \(2018, Sections 1.2 and 2.2.4\)](#) and [Little and Rubin \(2019, Chapter 1\)](#).

Plausible MCAR examples

- ▶ A batch of paper questionnaires is lost due to accidental water damage.
- ▶ A lab machine fails for one randomly chosen afternoon.

Probably not MCAR

- ▶ Low-income respondents are less likely to report income.
- ▶ Sicker patients are less likely to attend follow-up visits.
- ▶ Students with poor performance are more likely to skip an optional test.

In real datasets, MCAR should usually be treated as a claim that needs justification, not as the default assumption.

Under MCAR, complete-case analysis may be unbiased for many simple targets.

For estimating a population mean, if R is independent of Y ,

$$\mathbb{E}(Y \mid R = 1) = \mathbb{E}(Y).$$

So the mean among observed units targets the same population mean.

MCAR does not mean “no problem”. We still lose information (reducing sample size), so standard errors usually increase, reducing efficiency.

MCAR primarily protects against bias due to missingness; it does not protect against loss of precision.

Under MCAR, listwise deletion produces standard errors and significance levels that are correct for the reduced subset of data, but that are often larger relative to all available data

MCAR and complete-case analysis



What may be okay

- ▶ estimating simple means;
- ▶ estimating regression slopes when complete cases are representative;
- ▶ using standard likelihood methods on complete cases, with reduced sample size.

What still goes wrong

- ▶ lower effective sample size;
- ▶ larger standard errors;
- ▶ possible loss of power;
- ▶ wasted information from partially observed units.

Complete-case analysis is simplest under MCAR, but it might not be the most efficient use of the data.

Testing MCAR



Little's MCAR Test: (Little, 1988)

A formal statistical test to assess whether data are MCAR.

A significant result suggests deviation from MCAR.

Definition

The data are **missing at random** if, after conditioning on the observed data, missingness does not depend on the missing values:

$$\mathbb{P}(R \mid Y_{\text{obs}}, Y_{\text{mis}}, \psi) = \mathbb{P}(R \mid Y_{\text{obs}}, \psi).$$

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- ▶ it should not additionally depend on the values we failed to observe;
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MAR is weaker than MCAR, but still an assumption. This conditional definition follows [Rubin \(1976\)](#); see also [van Buuren \(2018, Sections 1.2 and 2.2.4\)](#).

Example: income nonresponse

Income is missing more often among older respondents and respondents from certain occupations. If age and occupation are observed, and after conditioning on them missingness does not depend on the actual income value, this is MAR.

Example: exam score

Exam score is missing more often for students with low attendance. If attendance is observed, and after conditioning on attendance missingness does not depend on the unobserved score, this is MAR.

MAR means the randomness in missingness can be explained by observed information.

MAR is conditional



A common misunderstanding:

“The missingness is related to age, so it is not random.”

Statistically, this can still be MAR if age is observed.

For example,

$$\mathbb{P}(R = 1 \mid Y_{\text{obs}}, Y_{\text{mis}}, \psi) = \mathbb{P}(R = 1 \mid \text{age}, \psi).$$

Another example: Women are (probably) less likely to disclose their weight, but their gender is recorded. In this case, weight is MAR.

The phrase “at random” in MAR is technical. It means conditionally random given the observed data.

Why MAR can still cause bias under deletion



Suppose low-attendance students are more likely to have missing exam scores.

If attendance is associated with exam score, then complete cases have a different attendance distribution from the full class.

Complete-case mean score estimates

$$\mathbb{E}(Y \mid R = 1),$$

not necessarily

$$\mathbb{E}(Y).$$

Under MAR, deletion can be biased if the observed variables related to missingness are also related to the outcome.

A small numerical MAR example



Group	True mean score	Fraction in class	Observation probability
High attendance	80	0.5	0.9
Low attendance	60	0.5	0.3

True class mean:

$$0.5(80) + 0.5(60) = 70.$$

Observed mean among complete cases:

$$\frac{0.5(0.9)(80) + 0.5(0.3)(60)}{0.5(0.9) + 0.5(0.3)} = 75.$$

Even though missingness depends only on observed group membership, complete-case analysis over-represents the high-attendance group.

the same bias principle under differential response is discussed in [Little and Rubin \(2019, Chapter 1\)](#) and [van Buuren \(2018, Section 1.2\)](#).

Typical approaches include:

- MAR is the main working assumption behind many standard missing-data methods.

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Definition

The data are **missing not at random** if the probability of missingness depends on the missing values even after conditioning on observed data:

$$\mathbb{P}(R \mid Y_{\text{obs}}, Y_{\text{mis}}, \psi) \text{ depends on } Y_{\text{mis}}.$$

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Intuition:

- ▶ the unobserved value helps determine whether it is observed;
- ▶ the missingness process is informative;
- ▶ the observed data alone cannot fully explain who is missing.

MNAR is also called **nonignorable missingness** in many texts (Rubin, 1976; Little and Rubin, 2019; Molenberghs and Kenward, 2007). See also van Buuren (2018, Sections 1.2 and 3.8).

- In MNAR settings, the missing part of the data is not just absent; it is systematically different in a way that the observed data may not reveal.

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A small numerical MNAR example



Let Y be binary:

$Y = 1$ means high risk, $Y = 0$ means low risk.

Assume

$$\mathbb{P}(Y = 1) = 0.5.$$

But suppose observation probabilities are

$$\mathbb{P}(R = 1 \mid Y = 1) = 0.2, \quad \mathbb{P}(R = 1 \mid Y = 0) = 0.8.$$

Then among observed cases,

$$\mathbb{P}(Y = 1 \mid R = 1) = \frac{0.5(0.2)}{0.5(0.2) + 0.5(0.8)} = 0.2.$$

The true high-risk fraction is 0.5, but the observed high-risk fraction is only 0.2.

This synthetic numerical example illustrates the selection-bias logic behind nonignorable mechanisms ([Little and Rubin, 2019](#); [Molenberghs and Kenward,](#)

Under MNAR, standard observed-data methods may fail unless the missingness process is explicitly modelled.

Possible strategies:

- ▶ collect more data on why values are missing;
- ▶ use subject-matter knowledge about the missingness process;
- ▶ specify a joint model for outcomes and response indicators;
- ▶ conduct sensitivity analyses over plausible MNAR mechanisms;
- ▶ report conclusions transparently as assumption-dependent.

MNAR assumptions are usually not identifiable from the observed data alone. They require modelling choices and sensitivity analysis.

MCAR, MAR, MNAR side by side



Type	Mechanism	Everyday interpretation
MCAR	$\mathbb{P}(R \mid Y_{obs}, Y_{mis}) = \mathbb{P}(R)$	Missingness unrelated to the data values.
MAR	$\mathbb{P}(R \mid Y_{obs}, Y_{mis}) = \mathbb{P}(R \mid Y_{obs})$	Missingness related only to observed information.
MNAR	depends on Y_{mis}	Missingness related to the unobserved value itself.

MCAR is a special case of MAR, but MAR is not the same as MCAR.

Which methods are naturally justified?



Type	Naive deletion?	Usually better approach
MCAR	Often unbiased, but inefficient	Use all available information; likelihood or imputation may improve precision.
MAR	Often biased unless special conditions hold	Observed-data likelihood, EM, multiple imputation, weighting.
MNAR	Generally risky	Model missingness, collect auxiliary information, sensitivity analysis.

How to classify an example



For any missing-data problem, ask three questions.

1. **What is missing?** Outcome, covariate, exposure, label, follow-up time, etc.
2. **What variables are observed?** Are predictors of missingness available in the dataset?
3. **Could missingness depend on the missing value itself?** Is there a plausible mechanism by which the unobserved value affects whether it is observed?

Classification depends on the data-collection story, not only on the data table.

1. A computer crash deletes a random 5% of lab measurements.
2. Income is missing more often for younger people; age is observed.
3. Income is missing more often for high-income individuals, even after conditioning on observed variables.
4. Blood pressure follow-up is missing more often for patients whose previous blood pressure was high.
5. A satisfaction score is missing because dissatisfied customers refuse to answer the survey.

Classroom classification exercise: answers



1. **MCAR:** a random computer crash is unrelated to either the observed or unobserved lab values.
2. **MAR:** missingness depends on age, and age is observed.
3. **MNAR:** missingness still depends on the unobserved income value after conditioning on observed variables.
4. **MAR:** missingness depends on previous blood pressure, which is observed, not necessarily on the unobserved follow-up value.
5. **MNAR:** dissatisfied customers refuse to answer because of the unreported satisfaction value itself.

These answers are model-based classifications: a different data-collection story or extra observed variables could change the classification.

A full probabilistic model may factor as

$$f(Y_{\text{obs}}, Y_{\text{mis}}, R \mid \theta, \psi) = f(Y_{\text{obs}}, Y_{\text{mis}} \mid \theta) \cdot \mathbb{P}(R \mid Y_{\text{obs}}, Y_{\text{mis}}, \psi).$$

The first factor is the "data" model.

The second factor is the missingness mechanism.

If we care only about θ , do we always need to model ψ ?

This is where the notion of **ignorability** appears.

Ignorability: the basic idea



For likelihood-based inference, the missingness mechanism is often called **ignorable** when two conditions hold ([Rubin, 1976](#); [Little and Rubin, 2019](#)):

1. the data are MAR;
2. the parameters θ and ψ are distinct, meaning the parameter space separates as a product space.

Then likelihood inference for θ can be based on

$$L_{\text{obs}}(\theta) = f(Y_{\text{obs}} \mid \theta),$$

without explicitly modelling $\mathbb{P}(R \mid Y, \psi)$. (For more details, see Theorem 7.1 of [Rubin \(1976\)](#)).

Ignorable does not mean unimportant. It means the missingness model can be omitted for a particular likelihood calculation under specific assumptions.

Do not confuse these two different ideas.

Listwise deletion

Throw away all incomplete records
and analyze only complete cases.

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$$L_{\text{obs}}(\theta) = f(Y_{\text{obs}} \mid \theta).$$

$$L_{\text{obs}}(\theta) = f(Y_{\text{obs}} \mid \theta).$$

The EM algorithm helps by alternating between:

- ▶ estimating the missing sufficient information conditionally on what was observed;
- ▶ maximizing an expected complete-data log-likelihood.

Can we test the missingness assumption?



Some aspects of missingness can be explored from the observed data.

Examples:

- ▶ compare observed covariates between complete and incomplete cases;
- ▶ regress the missingness indicator R on observed variables;
- ▶ visualize missingness patterns;
- ▶ use formal tests for MCAR in some multivariate settings, such as Little's test ([Little, 1988](#)).

Testing MAR vs MNAR



MNAR vs. MAR cannot be tested simply because data that are MNAR involve unobserved data. The deciding information is in Y_{mis} .

MAR itself is an unverifiable assumption, and must be justified in the context of each problem.

Practical workflow for missing data



1. Describe the amount and pattern of missingness.
2. Ask how each variable became missing.
3. Identify observed variables related to missingness.
4. Decide which assumption is plausible: MCAR, MAR, or MNAR.
5. Choose a method consistent with that assumption.
6. Report the assumption and, when needed, sensitivity analyses.

Bias and efficiency: two different concerns



Bias

Does the estimator target the correct population quantity?

Efficiency

How much uncertainty remains around the estimator?

Missing data can affect both:

- ▶ MCAR mainly causes loss of efficiency, though careless analysis can still waste information;
- ▶ MAR can cause bias under deletion but can often be handled by model-based methods;
- ▶ MNAR can cause bias unless the missingness mechanism is addressed.

“No bias” and “no loss of information” are not the same statement.

Why single imputation is not a full solution



A tempting approach is to fill in missing values once and then analyze the completed dataset.

Examples:

- ▶ replace missing income by mean income;
- ▶ replace missing score by predicted score from regression;
- ▶ fill missing lab value by the previous observed value.

Single imputation often treats imputed values as if they were truly observed. This usually understates uncertainty.

Multiple imputation was developed to reflect both sampling uncertainty and imputation uncertainty ([Schafer, 1997](#); [Carpenter and Kenward, 2013](#)).

Complete-case analysis: when to be suspicious



Be suspicious of complete-case analysis when:

- ▶ the missingness rate is large;
- ▶ missingness differs strongly by observed covariates;
- ▶ variables related to missingness are also related to the outcome;
- ▶ the missing variable is a key outcome or exposure;
- ▶ the reasons for nonresponse suggest discomfort, severity, risk, or cost;
- ▶ incomplete records still contain useful information.

Key Takeaway Messages



1. Missingness is described by both the observed data Y_{obs} and a response indicator R .
2. MCAR: missingness is unrelated to observed and missing values.
3. MAR: missingness may depend on observed values, but not on missing values after conditioning on what is observed.
4. MNAR: missingness depends on the unobserved value itself, even after conditioning on observed information.
5. Complete-case analysis is not automatically valid; its validity depends on the missingness mechanism and the target of inference.
6. MAR plus parameter distinctness gives ignorability for likelihood inference, not permission to throw away useful incomplete records.

Before choosing an imputation, likelihood, weighting, or EM-based method, first ask: why are these values missing?

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Stef van Buuren. *Flexible Imputation of Missing Data*. Chapman and Hall/CRC, Boca Raton, FL, 2 edition, 2018. doi: 10.1201/9780429492259. URL <https://stefvanbuuren.name/fimd/>.